# Genuine Multipartite Entanglement Analysis — Molecular System

## MOLECULE

NH2-  (NH2-)
Basis set    : cc-pVDZ
Charge       : -1
Multiplicity : 1
Total e-     : 10
n-body order : 3

## GROUND STATE ENERGIES

E0 (Ha)  : -55.48749269
E1 (Ha)  : -55.35926382
Gap (Ha) : 0.12822887
HF  (Ha) : -55.48003660

## CONVEX-ROOF METHODS

Methods run    : gcg
Restarts       : 8

C_LB  : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB  : eigendecomposition bound

## ACTIVE SPACE  CAS(6,5)

Frozen core orbs : 2
Frozen e-        : 4
Active e-        : 6
Active orbs      : 5
Spin-orbitals    : 10
Hilbert dim      : 100

## SPIN PROPERTIES

<S2>         : 0.000000
S            : 0.0000
<Sz>         : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

## GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

## DOUBLE OCCUPANCY

<D> total  : 2.992471
<D> / site : 0.598494

## THIS PDF

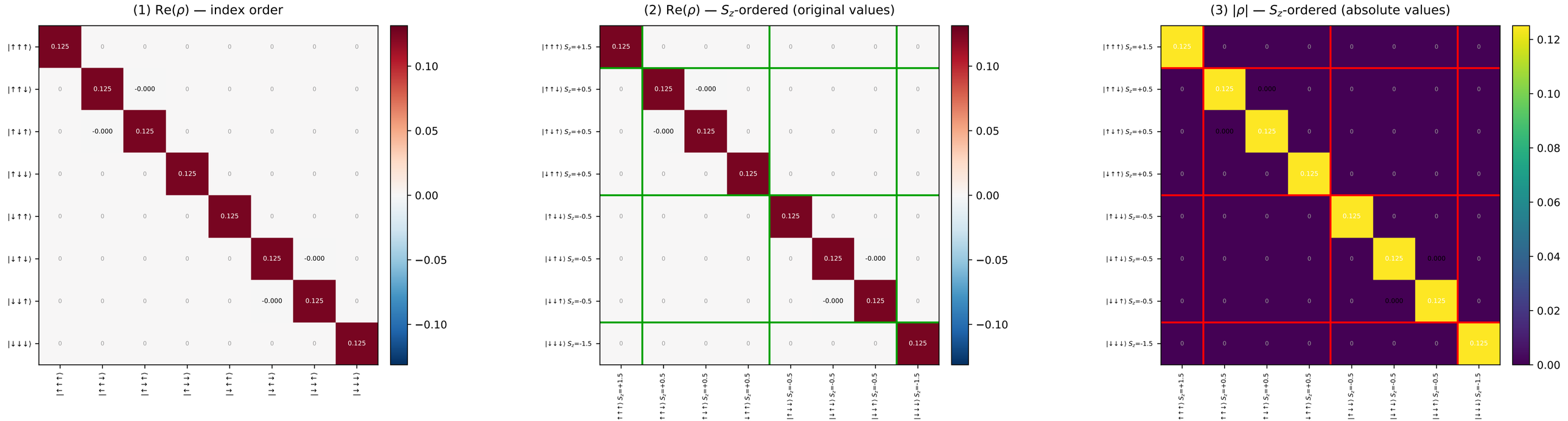
Body order : 3
Clusters   : 10

Date: 2026-07-15  09:31

## Summary — 3-Body RDMs  (NH2-, CAS(6,5))

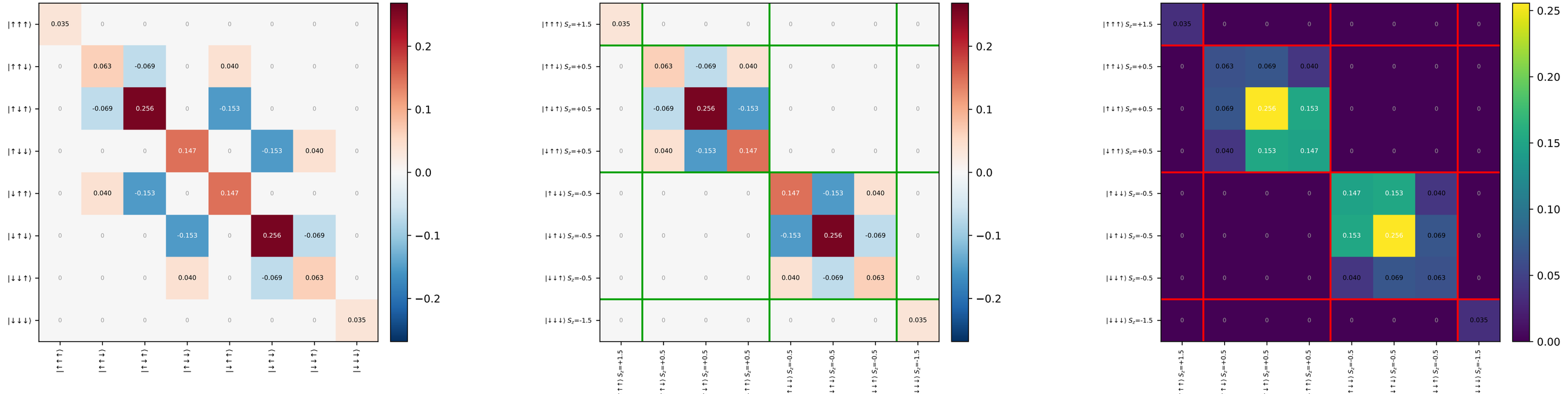
| Orbitals | Purity | Entropy | C_GME (pure) | C_LB | C_gcg | C_UB | State |
|---|---|---|---|---|---|---|---|
| 0,1,2 | 0.125000 | 2.079441 | — | — | — | — | STRONGLY MIXED |
| 0,1,3 | 0.302464 | 1.492167 | — | — | — | — | MIXED |
| 0,1,4 | 0.428599 | 1.040659 | — | — | — | — | MIXED |
| 0,2,3 | 0.130564 | 2.054887 | — | — | — | — | STRONGLY MIXED |
| 0,2,4 | 0.235332 | 1.713302 | — | — | — | — | MIXED |
| 0,3,4 | 0.495734 | 0.721811 | — | — | — | — | MIXED |
| 1,2,3 | 0.255909 | 1.646852 | — | — | — | — | MIXED |
| 1,2,4 | 0.126994 | 2.071023 | — | — | — | — | STRONGLY MIXED |
| 1,3,4 | 0.492355 | 0.744579 | — | — | — | — | MIXED |
| 2,3,4 | 0.226664 | 1.741217 | — | — | — | — | MIXED |

**3-Body RDM (Sites 0,1,2) — same matrix in all three views**

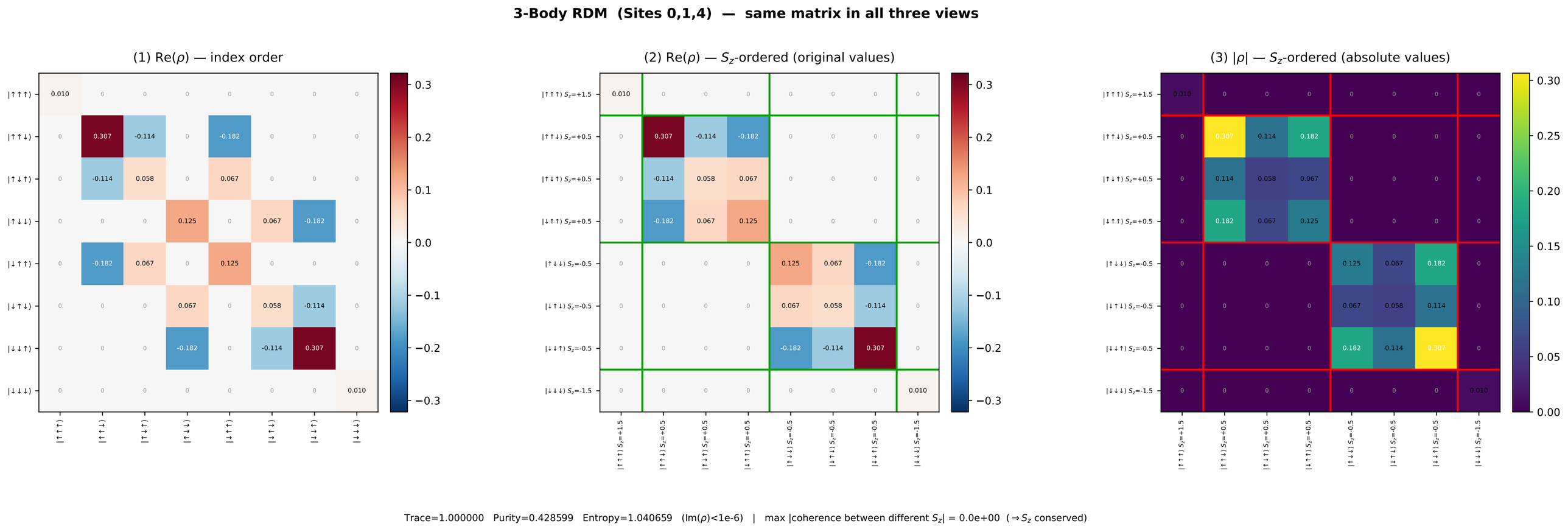



Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)

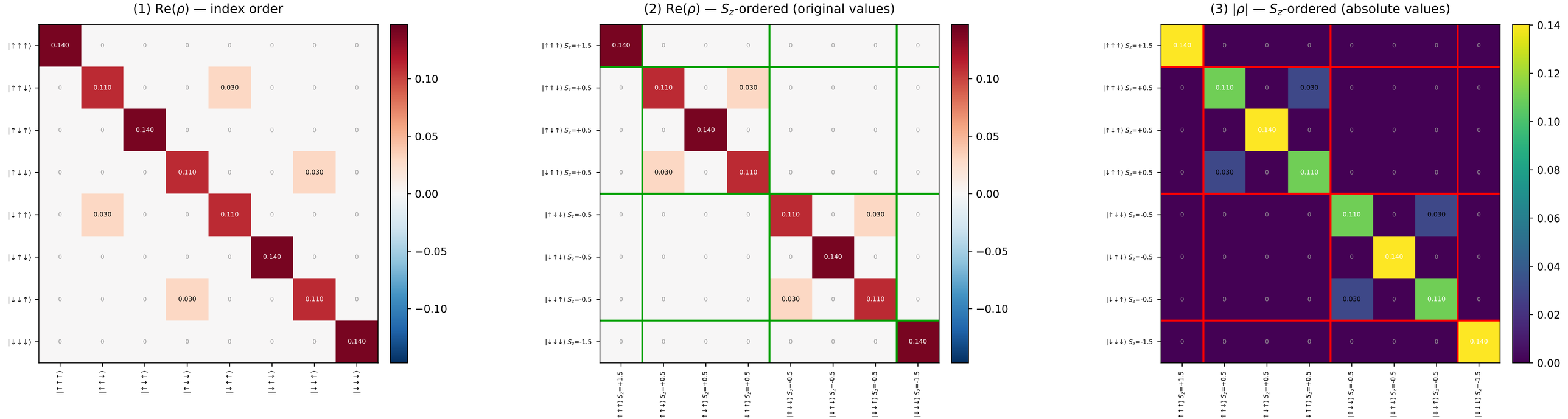
# 3-Body RDM (Sites 0,1,3) — same matrix in all three views




Trace=1.000000 Purity=0.302464 Entropy=1.492167 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)


3-Body RDM (Sites 0,1,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
Trace=1.000000 Purity=0.428599 Entropy=1.040659 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)

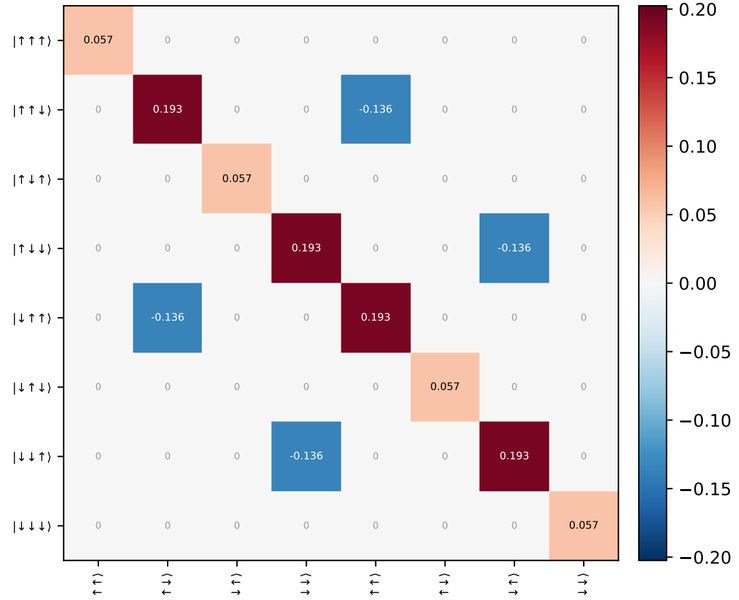
**3-Body RDM (Sites 0,2,3) — same matrix in all three views**



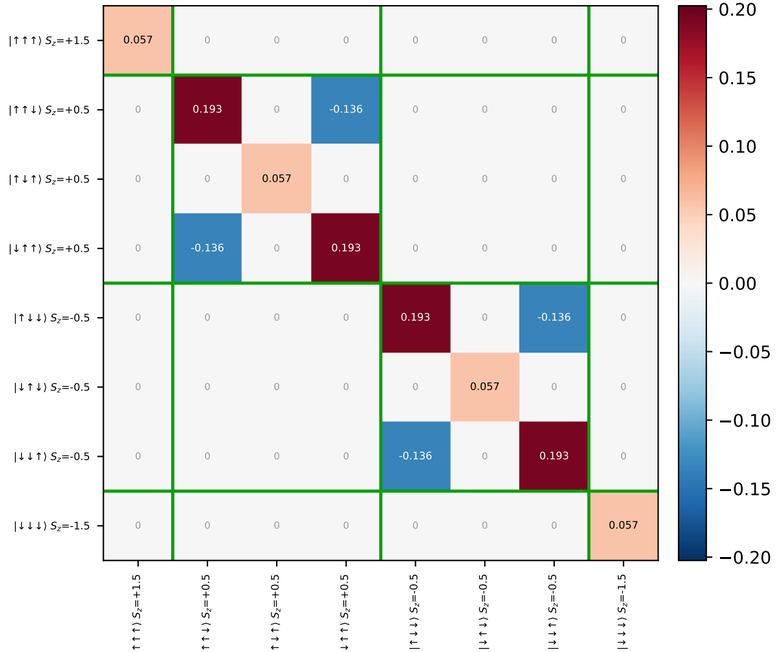

Trace=1.000000 Purity=0.130564 Entropy=2.054887 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)

**3-Body RDM (Sites 0,2,4) — same matrix in all three views**

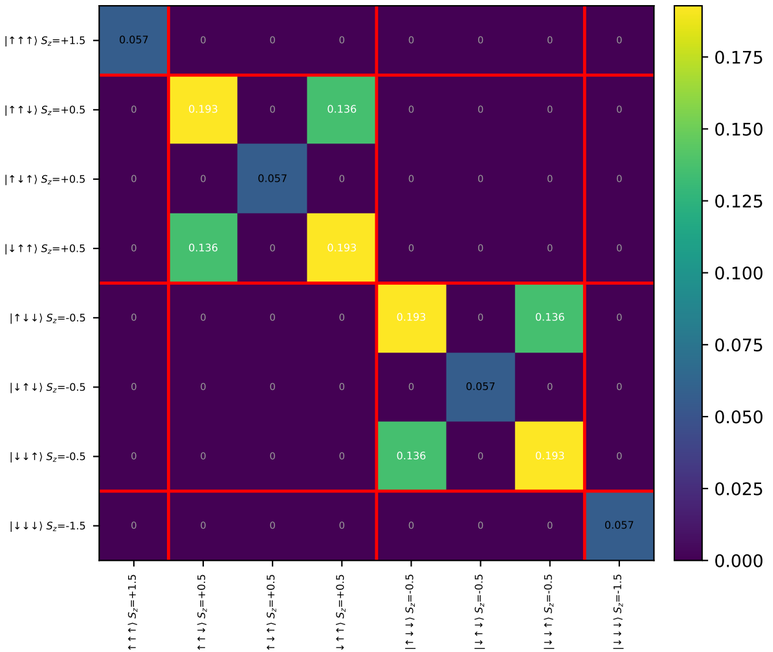
(1) Re($\rho$) — index order



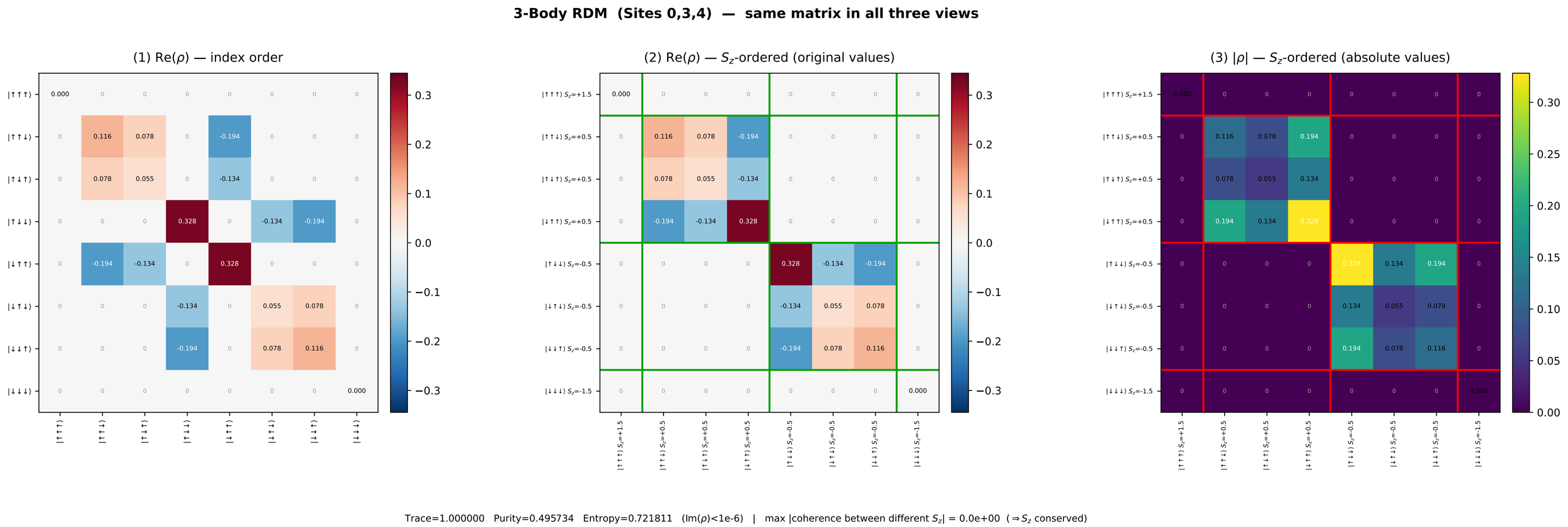

(2) Re($\rho$) — $S_z$-ordered (original values)

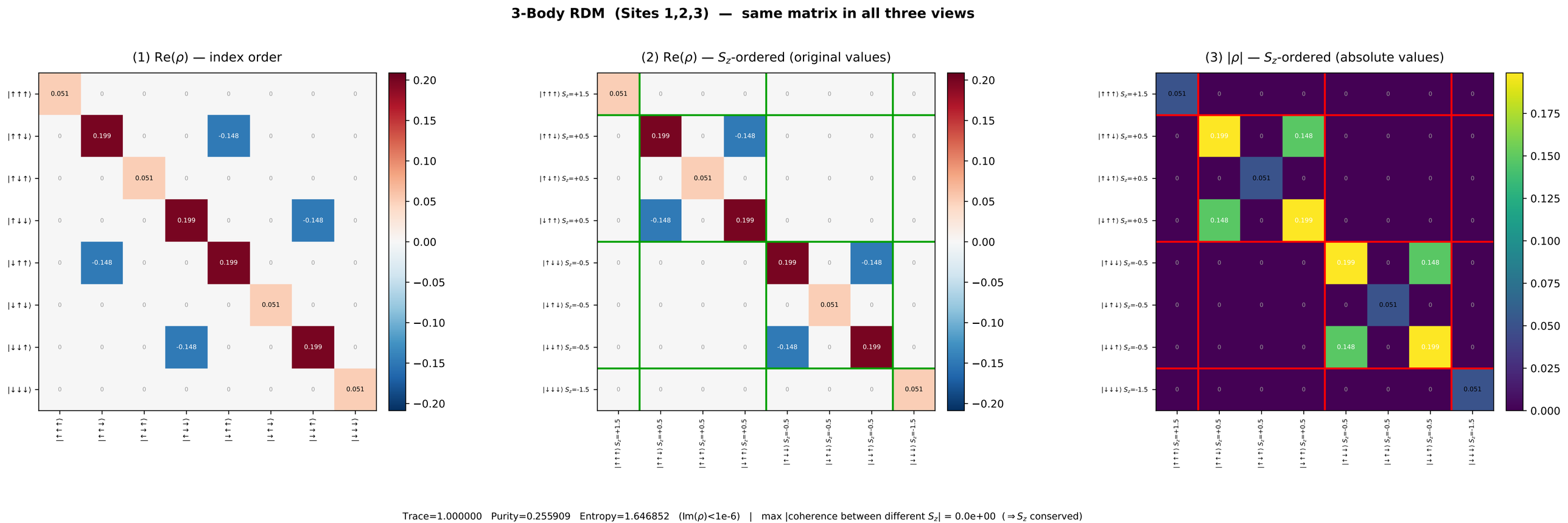


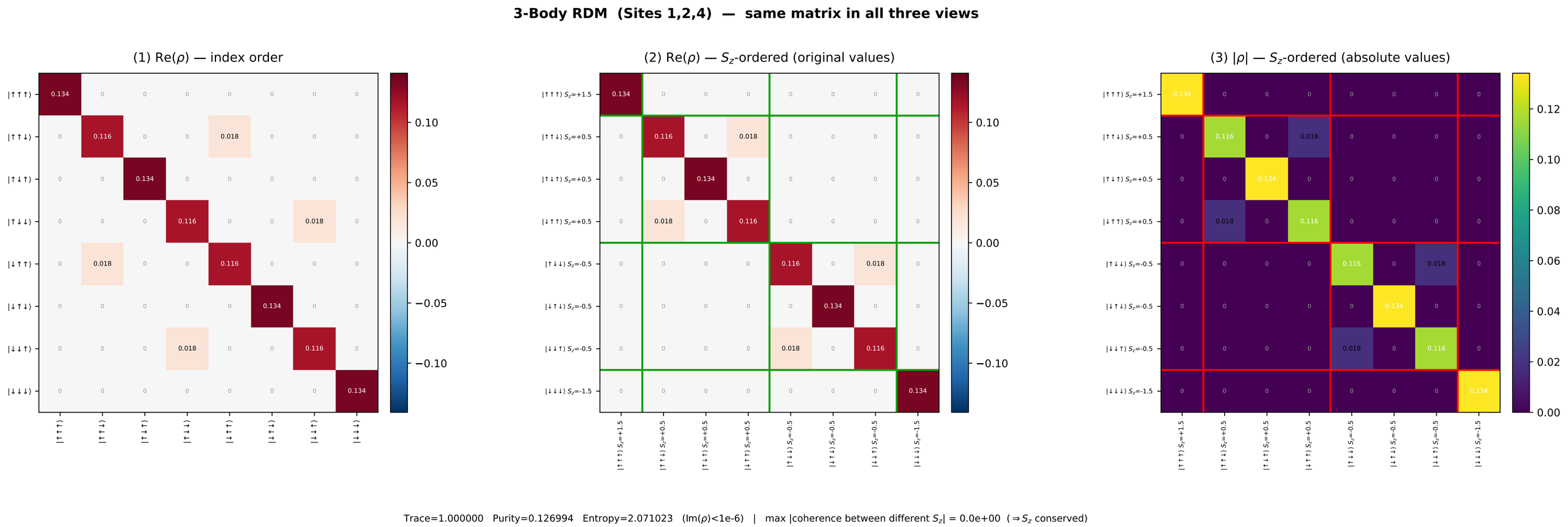

(3) $|\rho|$ — $S_z$-ordered (absolute values)



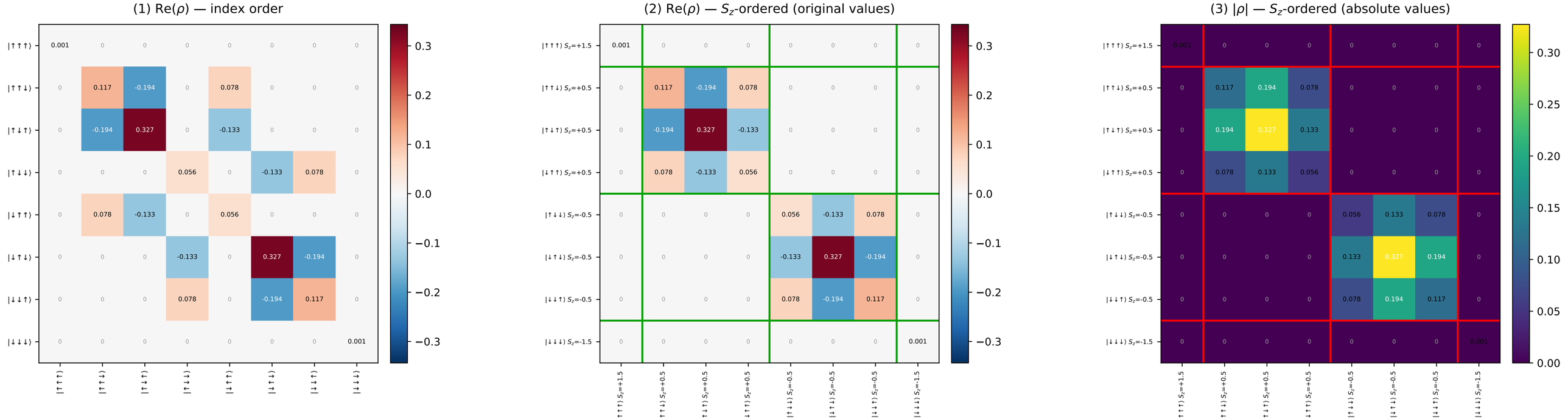

Trace=1.000000 Purity=0.235332 Entropy=1.713302 (Im($\rho$)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 ($\Rightarrow S_z$ conserved)


3-Body RDM (Sites 0,3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
Trace=1.000000 Purity=0.495734 Entropy=0.721811 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)


3-Body RDM (Sites 1,2,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
Trace=1.000000 Purity=0.255909 Entropy=1.646852 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)

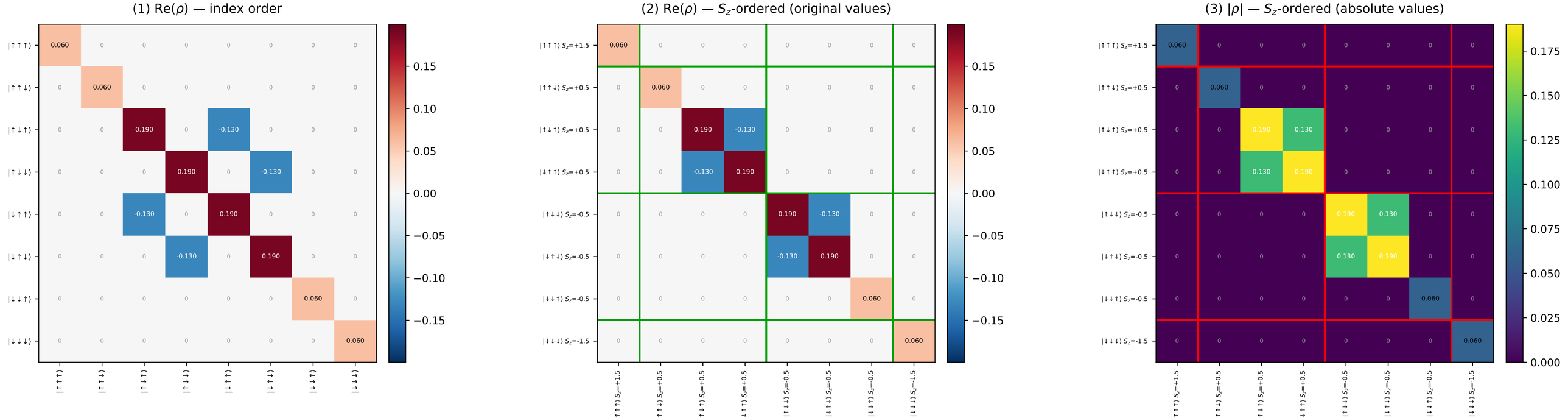

3-Body RDM (Sites 1,2,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
Trace=1.000000 Purity=0.126994 Entropy=2.071023 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)

# 3-Body RDM (Sites 1,3,4) — same matrix in all three views




Trace=1.000000  Purity=0.492355  Entropy=0.744579  (Im(ρ)<1e-6)  |  max |coherence between different $S_z$| = 0.0e+00  (⇒ $S_z$ conserved)

# 3-Body RDM (Sites 2,3,4) — same matrix in all three views




Trace=1.000000 Purity=0.226664 Entropy=1.741217 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)